# Highlights

**UL-TransXNet: A Compact Architecture for Multi-Dataset Ultrasound Lesion Classification**

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- UL-TransXNet classifies ultrasound lesions from ROI crops.
- Three public datasets support multi-dataset in-domain evaluation.
- Ablations isolate MCA, MUDD, and differential attention.
- Automatic ROI experiments connect localization and classification.
- Android experiments quantify accuracy–latency trade-offs.

# UL-TransXNet: A Compact Architecture for Multi-Dataset Ultrasound Lesion Classification

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**Abstract**

Ultrasound lesion classification is challenging because speckle noise, weak boundaries, low contrast, and acquisition-dependent variation obscure lesion cues. We present UL-TransXNet, a compact ROI-based TransXNet family architecture for benign–malignant ultrasound lesion classification. Lesion annotations are used to construct expanded ROI crops, after which a parameter/MAC-efficient hierarchical backbone models local texture and broader contextual relationships. We evaluate the model on three public datasets: TN5000 for thyroid nodules, BUSI for breast lesions, and AUL for liver lesions. Under dataset-specific training and the unified ROI reporting protocol, UL-TransXNet obtains mean test AUC/balanced-accuracy pairs of 0.9483/0.8726 on TN5000, 0.8998/0.7961 on BUSI, and 0.8767/0.8200 on AUL. The result does not imply uniform superiority: ConvNeXt-Tiny remains slightly stronger on TN5000 at the selected operating point. Instead, the main finding is multi-dataset in-domain competitiveness, with the strongest average performance across the three datasets and clear gains

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on BUSI and AUL. TN5000 ablations show that the TransXNet family refinements improve operating-point behavior, while MCA contributes mainly to thresholded decision quality and multi-dataset operating behavior rather than to a uniform AUC gain. Automatic-ROI experiments provide workflow evidence: TN5000 verifies detector-front-end feasibility on the primary benchmark, while BUSI/AUL compare oracle ROI, detector-predicted ROI, and full-image inputs and show that automatic crops improve over full-image classification but remain below oracle ROI performance. Android experiments quantify prototype inference trade-offs, where horizontal-flip test-time augmentation improves AUC from 0.9402 to 0.9466 at 226.2 ms per image. These results support UL-TransXNet as a competitive compact ultrasound classifier, while broader external generalization and clinical deployment remain open.

*Keywords:* Ultrasound image classification, Thyroid nodule, Breast ultrasound, Liver ultrasound, Compact neural network, Mobile inference prototype

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## 1. Introduction

Ultrasound image analysis has become an important pattern recognition problem because ultrasound is widely used in screening, diagnosis support, and follow-up examination for multiple organs. Compared with computed tomography and magnetic resonance imaging, ultrasound is inexpensive, radiation-free, portable, and suitable for repeated acquisition. These advantages make ultrasound attractive for computer-aided diagnosis, but they also make the recognition problem visually difficult. Speckle noise, low con-

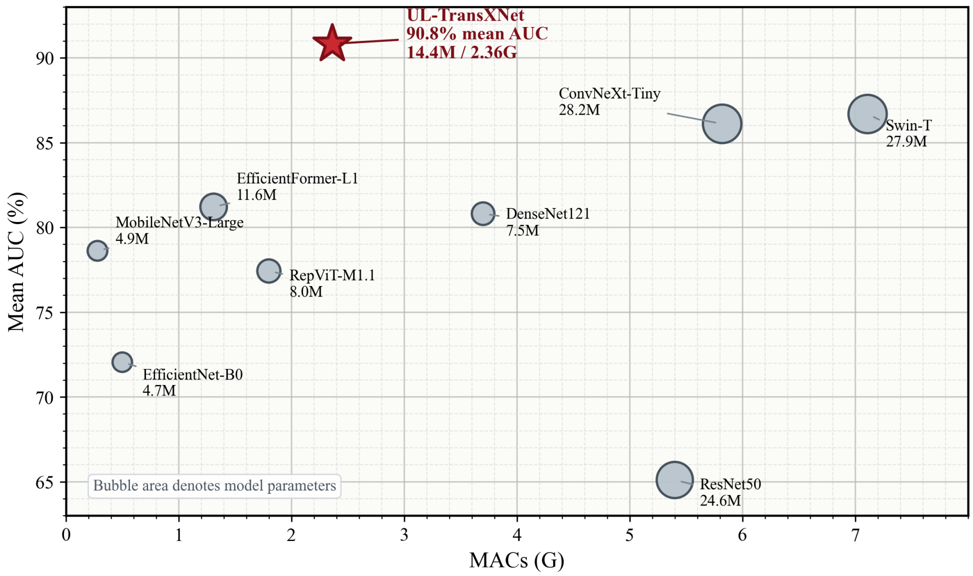


Figure 1: Accuracy–complexity trade-off on the three public ultrasound datasets. The vertical axis reports the mean AUC over TN5000, BUSI, and AUL; the horizontal axis reports MACs for one forward pass; bubble area denotes model parameters.

trast, blurred lesion boundaries, operator dependency, and device-specific appearance shifts can make benign–malignant discrimination substantially harder than standard natural-image classification. A useful ultrasound classifier therefore needs to preserve local lesion details, use surrounding tissue context, and remain compact enough for edge-side feasibility studies.

Public datasets now make it possible to evaluate ultrasound classifiers beyond a single organ. TN5000 provides a large public benchmark for thyroid nodule analysis [1]. BUSI is widely used for breast ultrasound lesion classification [2]. The Annotated Ultrasound Liver (AUL) dataset broadens the setting to liver ultrasound [3]. These datasets are not interchangeable: they differ in anatomy, acquisition appearance, class prevalence, annotation style, and split protocol. This makes them useful for testing whether an architectural idea is only tuned to one benchmark or remains competitive under different public ultrasound settings.

Recent deep learning methods have improved ultrasound analysis, but two gaps remain important for practical lesion-classification systems. First, many studies report strong results on one dataset while providing limited evidence that the same architecture remains useful under other public ultrasound benchmarks. Second, increasingly complex backbones can obscure the practical trade-off between accuracy and computational cost. This trade-off is not a secondary issue in ultrasound: many support scenarios require stable inference under constrained hardware and repeatable preprocessing.

This paper addresses these gaps with UL-TransXNet, a compact TransXNet family classifier for ROI ultrasound lesions. The study is deliberately organized as an evidence chain rather than as a broad claim of universal superiority. TN5000 is treated as the primary large benchmark. BUSI and AUL provide additional dataset-specific in-domain evaluations on breast and liver ultrasound. Structure and module ablations quantify how the TransXNet-family refinements behave under the unified protocol. A mobile prototype then examines whether the resulting model family can occupy a practical accuracy–latency region.

The main contributions are summarized as follows:

- We present UL-TransXNet, a compact ROI-based architecture for ultrasound lesion classification that combines hierarchical local–global feature modeling with a simple lesion-centered input protocol.
- We evaluate the model on TN5000, BUSI, and AUL, and we explicitly

separate multi-dataset in-domain competitiveness from stronger but unsupported claims of cross-organ generalization or uniform superiority.

- We provide unified-protocol TN5000 ablations that distinguish the effects of the structural progression, MCA, MUDD, and differential attention.
- We report Android measurements for a TN5000-calibrated prototype to show the speed–accuracy trade-off of the compact model family.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed method. Section 4 presents the datasets and protocol. Section 5 reports the main experimental findings. Section 6 discusses the implications and limitations, and Section 7 concludes the paper.

## 2. Related Work

### 2.1. Ultrasound lesion classification

Ultrasound lesion classification has been studied most extensively in breast and thyroid imaging, where public data availability has enabled repeated benchmarking [2, 1]. Earlier breast ultrasound work often relied on handcrafted morphological and texture descriptors combined with conventional classifiers [4]. More recent deep models have shifted the focus toward representation learning directly from lesion-centered images or regions of interest. For example, Luo *et al.* proposed a multi-view learning strategy for breast ultrasound tumor classification in *Pattern Recognition*, showing that task-specific view construction can improve classification accuracy [5]. These studies confirm that ultrasound lesion classification benefits from architectures that can retain subtle boundary, texture, and contextual cues. However, the majority of published systems remain strongly tied to a single organ or a single benchmark.

Compared with breast ultrasound, public thyroid and liver ultrasound classification benchmarks are less mature but increasingly important. TN5000 has substantially improved the benchmarking environment for thyroid nodule analysis by providing a large-scale public dataset and standardized evaluation protocol [1]. The AUL dataset broadens public benchmarking to liver ultrasound by offering annotated benign, malignant, and normal cases [3]. Taken together, these datasets suggest that a useful ultrasound classification architecture should not only perform well on one organ, but also remain competitive when transferred to datasets with different anatomy, lesion appearance, and acquisition characteristics.

### *2.2. Efficient visual architectures*

The design of efficient visual backbones has rapidly evolved from classical convolutional networks to compact CNN–Transformer hybrids. ResNet, DenseNet, EfficientNet, and MobileNetV3 established widely used convolutional design principles such as residual learning, dense feature reuse, compound scaling, and hardware-aware mobile optimization [6, 7, 8, 9]. Vision Transformer and DeiT then demonstrated the strength of attention-based image recognition, while Swin Transformer and ConvNeXt clarified the practical value of hierarchical representations and convolutional inductive biases [10, 11, 12, 13]. These models form the main reference space for modern visual backbones.

Recent lightweight designs focus less on raw parameter reduction and more on where computation should be allocated. MobileViT, EdgeNeXt, EfficientFormer, RepViT, and TransXNet are representative examples of this direction, combining local detail modeling, global context aggregation, and deployment-aware efficiency in different ways [14, 15, 16, 17, 18]. This line of work is particularly relevant to ultrasound analysis, where lesions often occupy a small portion of the image and subtle structural changes can be diagnostically important. Our architecture is built in this efficient-backbone context rather than as a detached medical-only network.

Attention refinements are also relevant to this design space. Differential Transformer introduced differential attention as the difference between two softmax attention maps to reduce attention noise and emphasize relevant context [19]. We do not treat this mechanism as a new primitive. Instead, we adapt it inside a compact ultrasound ROI classifier, where suppressing common speckle and background responses is useful for lesion-focused discrimination.

Medical image analysis has also begun to adopt lightweight hybrid designs. LightCM-PNet emphasizes real-time ultrasound segmentation under deployment constraints [20], while Lite-MixedNet explores efficient CNN–Transformer fusion for medical image segmentation [21]. These works show that lightweight design is becoming a first-class consideration rather than a secondary implementation detail. Our work follows this trend, but targets ultrasound lesion classification rather than segmentation. The goal is not to propose an entirely separate paradigm, but to derive a coherent and task-motivated architecture that preserves efficiency while improving fine-grained lesion discrimination.

### 2.3. Multi-dataset validation in medical image analysis

Reporting results on a single benchmark is often insufficient in medical image analysis because performance may depend strongly on dataset-specific factors such as annotation protocol, class prevalence, acquisition device, and institution-specific image characteristics [22, 23]. This issue is especially pronounced for ultrasound, where both image quality and lesion appearance can vary substantially across clinical settings. Recent medical-image-analysis studies have therefore begun to supplement their main benchmarks with robustness analysis, external validation, or deployment-oriented testing [24, 20]. Such evidence does not automatically prove universal generalization, but it gives a more credible picture of where a method is strong and where it remains fragile.

Our paper adopts this perspective. The primary claim is strong dataset-specific in-domain performance across several public ultrasound datasets, with TN5000 as the main benchmark and BUSI/AUL as additional public benchmarks on breast and liver ultrasound. BUSI and AUL help characterize performance under different anatomy, lesion appearance, and acquisition characteristics, but they are not treated as thyroid-to-breast or thyroid-to-liver external generalization tests. Temperature scaling is used only as validation-derived post-hoc calibration, following the established calibration literature [25]. This positioning keeps the evidence chain aligned with what the experiments can defensibly support.

## 3. Method

### 3.1. Problem formulation

Given an ultrasound region-of-interest (ROI) image $x$, the goal is to predict the lesion label $y \in \{0, 1\}$ for benign–malignant classification, where 0 denotes benign and 1 denotes malignant. The model outputs a two-dimensional logit vector $z = f_\theta(x)$, and the malignant posterior is computed as

$$p(y = 1 \mid x) = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)}. \tag{1}$$

The main paper uses the same binary formulation across TN5000, BUSI, and AUL, so that the proposed architecture can be evaluated under a unified lesion-classification setting.

### 3.2. ROI-based input construction

Figure 2 summarizes the input construction. The lesion annotation defines the initial bounding box. The box is expanded around the lesion center, clipped to the image boundary, and resized to the model input size. The crop preserves the expanded lesion aspect ratio before resizing; it is therefore an expanded rectangular ROI crop rather than a forced square crop. This procedure keeps the lesion centered while retaining a controlled amount of surrounding tissue context. The main classification protocol therefore uses annotation-derived ROI crops; closed-loop experiments in Section 4 evaluate detector-predicted ROI crops as an auxiliary inference-time front end.

### 3.3. Overall architecture

Figure 3 illustrates UL-TransXNet, the architecture used in the main experiments. A $256 \times 256$ ROI crop is processed by a lightweight four-stage hier-

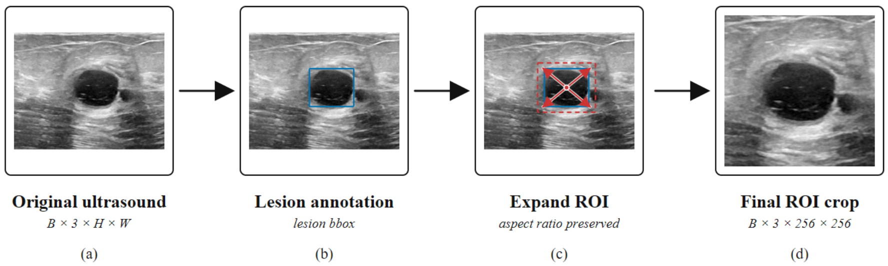


Figure 2: ROI-based input construction. The lesion annotation defines the initial bounding box; an expanded ROI crop centered on the lesion is extracted and resized to the model input size.

archical backbone. The backbone follows the tiny TransXNet family configuration with stage depths $[4, 4, 15, 4]$, embedding dimensions $[48, 96, 224, 448]$, head numbers $[1, 2, 4, 8]$, and spatial-reduction ratios $[8, 4, 2, 1]$. Each block keeps the central TransXNet idea of local–global token mixing: local convolutional dynamics preserve fine lesion structure, while the global branch models broader contextual relationships.

The backbone is used as a feature extractor. For the custom TransXNet-family models, the final backbone projection is instantiated as a trainable 1000-D feature projection before the task-specific binary head; it is not used as a frozen ImageNet-class logit vector. For timm baselines, the ImageNet classifier is removed and global-pooled features are passed to the same binary head family. The UL-TransXNet task head maps the 1000-D projection to the final binary prediction through a two-layer multilayer perceptron, $1000 \rightarrow 512 \rightarrow 2$, with GELU activation and dropout between the two linear layers. This keeps the classifier head simple and makes the backbone design the main

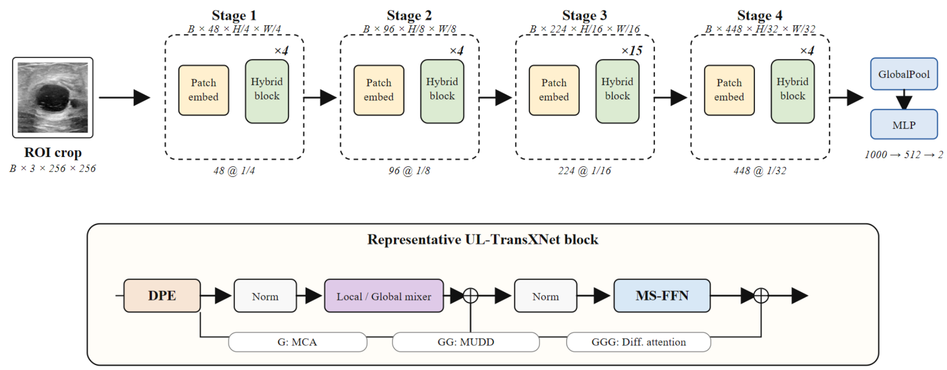


Figure 3: Overview of UL-TransXNet. The ROI crop is resized to $256 \times 256$ and processed by hierarchical patch/stem embedding, four TransXNet stages, global pooling, and a lightweight MLP head. The lower panel abstracts the representative block used to describe local/global token mixing, residual fusion, and MS-FFN refinement.

source of representation capacity.

### *3.4. Progressive TransXNet family refinements*

Rather than treating the final model as a one-shot design, we organize the architecture as a progressive refinement of the TransXNet baseline [18]. The baseline, denoted `TransXNet`, already combines dynamic local convolution and global attention in a compact hierarchical backbone. The ultrasound setting motivates additional refinements because lesion boundaries can be weak, discriminative textures can be local, and useful context can appear at different spatial scales.

The first refinement, `TransXNet-G`, enables multi-scale coordinate attention (MCA), inspired by coordinate attention for lightweight networks [26], in selected global-branch locations. The intent is to improve spatially aware feature selection without a large complexity increase. The second refinement, `TransXNet-GG`, adds multiway dynamic dense connections (MUDD) in deeper stages to encourage short-range feature reuse. The `TransXNet-GGG` variant introduces differential attention in the later global branch. We denote the main reported model as **UL-TransXNet**; in the ablation notation, it corresponds to the `TransXNet-GGG-MCA` configuration. This naming keeps the final model separate from the internal G/GG/GGG identifiers used for ablation. The model-family choice is audited with validation metrics across the dataset-specific training protocols in Section 5, rather than justified by a single TN5000 test optimum. The TN5000 ablations also show that the intermediate `TransXNet-GG` variant can be the strongest single TN5000 structure. This is an important part of the evidence rather than a contradiction: the refinements improve different operating characteristics, and MCA should be interpreted as an operating-point and multi-dataset behavior module rather than as a guaranteed AUC-improving switch.

### *3.5. Key modules*

*Multi-scale coordinate attention (MCA)..* When enabled, MCA operates on the global branch after channel splitting. It partitions channels into multiple scale groups, pools each group at a different spatial scale, injects coordinate-aware responses, and generates attention weights for spatially selective feature recalibration. In ultrasound images, lesions may differ through both coarse shape and fine echo patterns; MCA is intended to retain this scale diversity while keeping attention lightweight.

*Multiway dynamic dense connection (MUDD)..* MUDD is inserted in deeper stages, where semantic features are richer but optimization can be harder. The block feature is divided into multiple channel paths, and a lightweight weight generator computes path-wise aggregation weights from the current activation. Previous block features from the same stage are dynamically fused through these paths. This gives the deeper stages a compact adaptive memory without turning the backbone into a heavy dense network.

*Differential attention..* The `GGG` variant adapts differential attention from the Differential Transformer formulation [19], itself building on standard Transformer attention [27]. In selected later-stage global-branch locations, queries and keys are split into two groups and two attention maps are computed:

$$A_{\text{diff}} = \text{softmax}\left(Q_1 K_1^\top / \sqrt{d}\right) - \lambda \text{softmax}\left(Q_2 K_2^\top / \sqrt{d}\right), \tag{2}$$

where $\lambda$ is a learnable balancing coefficient. This module is therefore not claimed as a new attention primitive. In UL-TransXNet, it is used as an ultrasound-specific adaptation inside the global branch to suppress common background or speckle responses and sharpen lesion-related relational contrast when benign and malignant lesions share broad appearance statistics but differ in subtle internal organization.

These modules are introduced as a progressive design family, not as a full factorial decomposition. The ablation study therefore reports both structure-level variants and core-module variants to make the interpretation explicit.

### *3.6. Complexity analysis*

Table 1 reports the complexity of the comparison pool and the TransXNet family variants. The numbers are measured with the same binary-

Table 1: Model complexity of the comparison pool and TransXNet family variants. Parameters and MACs are measured for the full binary classifier, including the lightweight MLP head. MACs are reported for one forward pass at the listed input size.

| Method | Role | Input | Params (M) | MACs (G) |
|---|---|---|---|---|
| ResNet50 | Prior baseline | $256 \times 256$ | 24.56 | 5.40 |
| DenseNet121 | Prior baseline | $256 \times 256$ | 7.48 | 3.70 |
| EfficientNet-B0 | Prior baseline | $256 \times 256$ | 4.66 | 0.50 |
| MobileNetV3-Large | Prior baseline | $256 \times 256$ | 4.86 | 0.28 |
| Swin-T | Prior baseline | $256 \times 256$ | 27.91 | 7.11 |
| ConvNeXt-Tiny | Prior baseline | $256 \times 256$ | 28.21 | 5.82 |
| RepViT-M1.1 | Prior baseline | $256 \times 256$ | 8.04 | 1.80 |
| EfficientFormer-L1 | Prior baseline | $224 \times 224$ | 11.62 | 1.31 |
| TransXNet | Ours / ablation | $256 \times 256$ | 13.35 | 2.32 |
| TransXNet-G | Ours / ablation | $256 \times 256$ | 13.42 | 2.34 |
| TransXNet-GG | Ours / ablation | $256 \times 256$ | 14.15 | 2.38 |
| TransXNet-GGG-noMCA | Ours / ablation | $256 \times 256$ | 14.29 | 2.34 |
| **UL-TransXNet** | **Ours** | $256 \times 256$ | 14.37 | 2.36 |

classification wrapper used by the training scripts, so they include both the backbone and the lightweight MLP head. All models use $256\times256$ ROI inputs except EfficientFormer-L1, which follows its $224\times224$ setting. UL-TransXNet is not the smallest model and is not optimized for local CUDA latency, but it uses fewer parameters and MACs than the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T while preserving stronger multi-dataset in-domain behavior on BUSI and AUL.

Table 2: Overview of the main datasets used in the paper. BUSI and AUL are used in a benign–malignant lesion-classification setting.

| Dataset | Organ | Images | B / M | Split protocol | Role |
|---|---|---|---|---|---|
| TN5000 | Thyroid | 5000 | 1426 / 3574 | Official 3500/500/1000 train/val/test | Main |
| BUSI | Breast | 647 | 437 / 210 | Fixed test 129; five training folds on trainval 518 | Additional |
| AUL | Liver | 635 | 200 / 435 | Fixed test 127; five training folds on trainval 508 | Additional |

## 4. Datasets and Experimental Setup

### *4.1. Datasets*

The main paper uses three public ultrasound datasets: TN5000 for thyroid nodules [1], BUSI for breast lesions [2], and the Annotated Ultrasound Liver (AUL) dataset for liver lesions [3]. TN5000 is the largest benchmark and is used for the most detailed ablation analysis. BUSI and AUL are used to test whether the architecture remains competitive beyond thyroid ultrasound.

### *4.2. Implementation details*

All datasets are processed in an ROI-based manner, but the crop expansion differs across benchmarks. TN5000 and BUSI use a lesion expansion ratio of 0.30, whereas AUL uses a lighter expansion ratio of 0.20. The lesion bounding box is read from XML or manifest metadata, expanded with a minimum crop size of 64 pixels, clipped to the image boundary, and resized to $256 \times 256$; when a reliable crop is unavailable, the whole image is used as a fallback. The current protocol preserves the expanded box aspect ratio before resizing and does not force a square crop. The only input-size exception in the comparison pool is EfficientFormer-L1, which is evaluated at $224 \times 224$

because its wrapper is not compatible with the $256 \times 256$ ROI setting used by the other models.

The training pipeline uses the same recipe family across the main runs: random horizontal flip, random rotation ($\pm 4°$), mild affine translation and scaling, brightness/contrast jitter, and random Gaussian blur. These perturbations are kept mild because ultrasound gray-scale intensity, speckle, and gain variation are part of the imaging domain rather than arbitrary color content. Images are loaded as RGB tensors for compatibility with ImageNet-initialized backbones; gray-scale inputs are converted to three channels by the image loader. Evaluation uses deterministic resizing and ImageNet normalization only. All models use AdamW with weight decay $10^{-4}$, a lightweight MLP head with dropout 0.3, and label smoothing 0.0. The learning-rate schedule is dataset-specific.

TN5000 uses the official train/val/test split and is reported with three random seeds (17, 27, and 37). The mainline setting uses 70 epochs, patience 12, manual class weights, and differential learning rates of $5 \times 10^{-5}$ for the backbone and $1.5 \times 10^{-4}$ for the classification head. BUSI uses a fixed held-out test set together with five training folds on the trainval subset; its main runs use 200 epochs, patience 36, class weighting, and the same $5 \times 10^{-5}/1.5 \times 10^{-4}$ learning-rate pair. AUL likewise uses a fixed held-out test set plus five training folds on trainval, but uses a lighter budget: 20 epochs, patience 4, no class weighting, top-2 checkpoint ensembling, and lower differential learning rates of $1 \times 10^{-5}$ for the backbone and $3 \times 10^{-5}$ for the classification head. Thus, BUSI/AUL mean $\pm$ standard deviation values quantify variation across training folds evaluated on a fixed test set, not an outer five-fold cross-validation estimate where each case appears in a test fold exactly once. The mainline scripts enable exponential moving average (EMA) with decay 0.9995; the exception in the baseline pool is BUSI RepViT-M1.1, for which EMA is disabled because the EMA variant produced unstable folds during verification.

For inference-time calibration, each run uses validation-derived temperature scaling [25], averages the top-$k$ validation-improved checkpoints (top-3 on TN5000 and BUSI, top-2 on AUL), applies horizontal-flip test-time augmentation, fits a scalar temperature on validation logits, and searches a decision threshold from 0.10 to 0.95 with step 0.01 using balanced accuracy as the default selection criterion. The resulting validation threshold is applied to the test predictions of that run. Main benchmark tables report mean $\pm$ standard deviation across the three TN5000 seeds or the five BUSI/AUL folds.

Balanced accuracy is used for threshold selection because it gives equal weight to benign and malignant recall and is less tied to class prevalence than raw accuracy. AUC is kept as the threshold-free ranking metric, while macro F1 and accuracy are reported as complementary operating-point summaries.

Importantly, all temperatures and decision thresholds are fixed strictly from validation-derived predictions. No test labels are used for temperature fitting, threshold tuning, or operating-point selection; the test set is used only once for final reporting after these choices are fixed.

### *4.3. Automatic ROI closed-loop protocol*

The main experiments use annotation-derived ROI crops to isolate classifier behavior under a controlled input protocol. To evaluate whether this ROI classifier can be connected to an automatic front end at inference time, we additionally train one-class lesion detectors on BUSI and AUL and perform closed-loop classification with detector-predicted crops. The detector is used only as an auxiliary ROI generator; it is not presented as a novel detection method.

For each BUSI/AUL fold, a one-class YOLO11n detector is trained on the corresponding training annotations and evaluated on the held-out test images. A parallel TN5000 automatic-ROI check trains YOLO11n detectors under three detector seeds and evaluates the official test split. The predicted lesion box with the highest confidence is exported into a VOC-style annotation file and passed through the same expansion, resize, normalization, temperature-scaling, and thresholding code used by the ROI classifier. We then compare three test-time inputs using the same trained UL-TransXNet checkpoints: *oracle ROI*, which uses the ground-truth lesion annotation; *automatic ROI*, which uses the detector-predicted box; and *full image*, which disables the lesion crop and resizes the complete ultrasound image. Thus, the closed-loop experiment tests whether detector-predicted crops can partially substitute annotation-derived ROI construction without retraining the classifier.

### *4.4. Compared methods*

The benchmark pool combines representative CNN, lightweight hybrid, and modern efficient baselines. The comparison set includes ResNet50 [6], EfficientNet-B0 [8], MobileNetV3-Large [9], Swin-T [12], DenseNet121 [7], ConvNeXt-Tiny [13], RepViT-M1.1 [17], and EfficientFormer-L1 [16]. The main paper table reports *Ours* together with the strongest non-*Ours* baseline on each dataset, where "strongest" is determined by mean test AUC under the same reporting convention. The full baseline table is provided in Appendix Table A.17.

### 4.5. Evaluation metrics and statistical analysis

The main metrics are AUC, balanced accuracy (BalAcc), macro F1, and accuracy. Unless explicitly stated otherwise, "F1" in the manuscript refers to macro F1. Because the tables emphasize repeated-seed/fold stability, results are reported as mean $\pm$ standard deviation. Thresholded metrics should be interpreted as validation-selected operating points rather than test-tuned oracle points.

### 4.6. Local efficiency profiling

Model efficiency is profiled on the TN5000 test split, which provides the largest held-out set among the main datasets. Local CUDA latency is measured on an NVIDIA GeForce RTX 4080 SUPER using batch size 1. Each model is instantiated with the same binary-classification wrapper used for complexity profiling, warmed up for 50 forward passes, and then timed for 300 model-only forward passes with CUDA synchronization. We report the mean model-only latency and the corresponding FPS. A separate TN5000 pipeline pass over 1000 test ROI crops is also logged for diagnosis, but the main trade-off table uses model-only latency to avoid mixing architecture cost with image decoding and CPU preprocessing overhead.

### 4.7. Mobile prototype protocol

Mobile inference is evaluated with an Android prototype built on ONNX Runtime 1.24.3. The reported batch experiments were executed on a Xiaomi device (model 24129PN74C, Android 16, arm64-v8a). The prototype study uses TN5000-calibrated UL-TransXNet ONNX exports and compares three inference variants: a single checkpoint, the same checkpoint with horizontal-flip test-time augmentation (TTA), and a three-checkpoint ensemble with TTA. Cold-start runs explicitly cleared the session cache before evaluation, whereas warm-start runs reused the loaded model session. The single-checkpoint mode was measured with one cold-start and three warm-start runs; the TTA and ensemble variants were measured with one cold-start and one warm-start run each. The prototype configuration used lesion expansion ratio 0.30, a fitted temperature of 1.819830, and decision threshold 0.73 inherited from TN5000 validation calibration. We report average end-to-end latency and the corresponding 95th percentile latency over the full ROI preprocessing-plus-inference pipeline.

## 5. Results

### 5.1. Main comparison results on public ultrasound datasets

Table 3 summarizes the main comparison results, reported as mean ± standard deviation across TN5000 seeds or BUSI/AUL training folds evaluated on the fixed held-out test sets. The evidence should be read carefully. On TN5000, ConvNeXt-Tiny is slightly stronger than the proposed model at the selected operating point. On BUSI and AUL, however, UL-TransXNet provides a clear gain over the strongest non-*Ours* baseline. Averaged over the three dataset-specific evaluations, the proposed model gives the strongest overall performance, especially in balanced accuracy.

These results delimit the central claim. The proposed model should not

Table 3: Main comparison results on the public ultrasound datasets. For each dataset, the table reports the proposed model and the strongest non-*Ours* baseline ranked by mean test AUC. F1 denotes macro F1.

| Dataset | Method | AUC | BalAcc | F1-macro | Acc |
|---|---|---|---|---|---|
| TN5000 | UL-TransXNet | 0.9483 ± 0.0006 | 0.8726 ± 0.0046 | 0.8503 ± 0.0123 | 0.8790 ± 0.0140 |
| TN5000 | ConvNeXt-Tiny | **0.9508 ± 0.0020** | **0.8794 ± 0.0023** | **0.8731 ± 0.0066** | **0.8957 ± 0.0065** |
| BUSI | UL-TransXNet | **0.8998 ± 0.0116** | **0.7961 ± 0.0104** | **0.8018 ± 0.0119** | **0.8140 ± 0.0130** |
| BUSI | Swin-T | 0.8190 ± 0.0091 | 0.7137 ± 0.0095 | 0.6896 ± 0.0300 | 0.7194 ± 0.0445 |
| AUL | UL-TransXNet | **0.8767 ± 0.0009** | **0.8200 ± 0.0487** | 0.7462 ± 0.0405 | 0.7685 ± 0.0418 |
| AUL | Swin-T | 0.8319 ± 0.0117 | 0.7783 ± 0.0325 | **0.7589 ± 0.0243** | **0.7717 ± 0.0211** |

be described as uniformly better than every baseline on every dataset. Its advantage is a multi-dataset in-domain pattern: it is competitive on TN5000 and substantially stronger on BUSI and AUL under dataset-specific training. The mean AUC over TN5000, BUSI, and AUL is 0.9083 for UL-TransXNet, compared with 0.8672 for the strongest-per-dataset non-*Ours* rows listed in Table 3. The corresponding mean balanced accuracy is 0.8296 versus 0.7905. This supports the claim that the architecture is a strong compact multi-dataset ultrasound classifier, not a claim of uniform dominance on the primary thyroid benchmark or external cross-organ generalization.

AUL illustrates the distinction between threshold-free ranking and thresholded operating-point metrics. UL-TransXNet has higher AUC and balanced accuracy than Swin-T, whereas Swin-T has slightly higher macro F1 and raw accuracy at its validation-selected threshold. We therefore interpret the AUL result as better ranking and class-balanced operating behavior, not as dominance over every thresholded metric.

Table 5 adds an offline paired statistical check over the same repeated

Table 4: Original TransXNet versus UL-TransXNet under the same dataset-specific reporting protocol. $\Delta$ denotes UL-TransXNet minus TransXNet.

| Dataset | TransXNet | | UL-TransXNet | | $\Delta$ | |
|---|---|---|---|---|---|---|
| | AUC | BalAcc | AUC | BalAcc | AUC | BalAcc |
| TN5000 | 0.9495 | 0.8697 | 0.9483 | 0.8726 | -0.0012 | 0.0028 |
| BUSI | 0.7227 | 0.6560 | 0.8998 | 0.7961 | 0.1770 | 0.1401 |
| AUL | 0.6582 | 0.6263 | 0.8767 | 0.8200 | 0.2186 | 0.1937 |

runs used by Table 3. On TN5000, the proposed model is slightly below ConvNeXt-Tiny. On BUSI and AUL, the paired differences are consistently positive for both AUC and balanced accuracy. Because the available repeated units are only three TN5000 seeds and five BUSI/AUL folds, these tests are used as stability evidence rather than as large-sample clinical significance claims.

Table 4 directly compares the original TransXNet baseline with UL-TransXNet under the same dataset-specific reporting protocol. This comparison is important because the method is a TransXNet-family adaptation rather than a detached backbone. The final model is not a TN5000 AUC improvement over the original TransXNet, but it slightly improves the TN5000 validation-selected operating point and gives much larger gains on BUSI and AUL.

Table 6 gives the corresponding complexity–performance view for the full comparison pool. This table makes the trade-off more explicit than the accuracy-only table: UL-TransXNet is parameter/MAC-efficient, but it is not the smallest model and is not CUDA-latency optimal. It achieves the

Table 5: Offline paired statistical comparison between UL-TransXNet and the strongest non-*Ours* baseline in Table 3. $\Delta$ denotes UL-TransXNet minus baseline. Confidence intervals are paired bootstrap 95% intervals over seeds or folds; $p$ is an exact two-sided paired sign-flip test. Because only three TN5000 seeds and five BUSI/AUL folds are available, these tests are intended as repeated-run stability checks rather than large-sample clinical significance tests.

| Dataset | Baseline | Metric | $n$ | $\Delta$ | 95% CI | $p$ |
|---|---|---|---|---|---|---|
| TN5000 | ConvNeXt-Tiny | AUC | 3 | -0.0025 | [-0.0040, -0.0006] | 0.2500 |
| TN5000 | ConvNeXt-Tiny | BalAcc | 3 | -0.0068 | [-0.0104, -0.0022] | 0.2500 |
| BUSI | Swin-T | AUC | 5 | +0.0807 | [0.0697, 0.0918] | 0.0625 |
| BUSI | Swin-T | BalAcc | 5 | +0.0823 | [0.0647, 0.0961] | 0.0625 |
| AUL | Swin-T | AUC | 5 | +0.0449 | [0.0375, 0.0574] | 0.0625 |
| AUL | Swin-T | BalAcc | 5 | +0.0417 | [0.0160, 0.0674] | 0.0625 |

highest mean AUC and mean balanced accuracy with 14.37M parameters and 2.36 GMACs, while its local CUDA batch-1 latency is 30.23 ms, slower than several GPU-optimized baselines. In contrast, the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T require 28.21M/5.82G and 27.91M/7.11G, respectively. The input column is included because EfficientFormer-L1 is evaluated at its native 224 input while the other models use the 256 ROI input used by the proposed model.

### 5.2. TN5000 structure ablation

Table 7 reports the unified-protocol TN5000 structure ablation. The strongest TN5000 structure is `TransXNet-GG`, which achieves the highest AUC, balanced accuracy, macro F1, and accuracy in this ablation group. The

Table 6: Complexity–performance trade-off of the comparison pool. CUDA latency is the mean model-only batch-1 latency measured on the local workstation; AUC values are mean test AUCs across TN5000 seeds or BUSI/AUL folds. Mean AUC and Mean BalAcc are averaged over the three datasets.

| Method | Type | Input | Params (M)↓ | MACs (G)↓ | CUDA ms↓ | CUDA FPS↑ | AUC↑ | | | Mean AUC↑ | Mean BalAcc↑ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | TN5000 | BUSI | AUL | | |
| MobileNetV3-Large | Mobile CNN | 256 × 256 | 4.86 | 0.28 | 3.17 | 315.1 | 0.8282 | 0.7947 | 0.7357 | 0.7862 | 0.7183 |
| EfficientNet-B0 | Efficient CNN | 256 × 256 | 4.66 | 0.50 | 6.53 | 153.1 | 0.7856 | 0.7659 | 0.6099 | 0.7205 | 0.6682 |
| EfficientFormer-L1 | Hybrid | 224 × 224 | 11.62 | 1.31 | 3.41 | 293.0 | 0.9249 | 0.7917 | 0.7196 | 0.8121 | 0.7330 |
| RepViT-M1.1 | Mobile CNN | 256 × 256 | 8.04 | 1.80 | 5.19 | 192.5 | 0.8403 | 0.8028 | 0.6797 | 0.7743 | 0.7020 |
| **UL-TransXNet** | **Ours** | 256 × 256 | 14.37 | 2.36 | 30.23 | 33.1 | 0.9483 | **0.8998** | **0.8767** | **0.9083** | **0.8296** |
| DenseNet121 | CNN | 256 × 256 | 7.48 | 3.70 | 7.19 | 139.1 | 0.9360 | 0.8154 | 0.6729 | 0.8081 | 0.7401 |
| ResNet50 | CNN | 256 × 256 | 24.56 | 5.40 | 3.25 | 307.5 | 0.6531 | 0.7519 | 0.5481 | 0.6510 | 0.6190 |
| ConvNeXt-Tiny | ConvNet | 256 × 256 | 28.21 | 5.82 | 3.30 | 303.3 | **0.9508** | 0.8050 | 0.8274 | 0.8611 | 0.7651 |
| Swin-T | Transformer | 256 × 256 | 27.91 | 7.11 | 6.32 | 158.3 | 0.9497 | 0.8190 | 0.8319 | 0.8669 | 0.7901 |

Table 7: Unified-protocol structure ablation on TN5000. Mean and standard deviation are computed over three seeds.

| Variant | AUC | BalAcc | F1-macro | Acc |
|---|---|---|---|---|
| TransXNet | 0.9495 ± 0.0020 | 0.8697 ± 0.0100 | 0.8366 ± 0.0222 | 0.8640 ± 0.0230 |
| TransXNet-G | 0.9481 ± 0.0013 | 0.8745 ± 0.0074 | 0.8395 ± 0.0220 | 0.8660 ± 0.0233 |
| TransXNet-GG | **0.9517 ± 0.0011** | **0.8784 ± 0.0076** | **0.8552 ± 0.0173** | **0.8827 ± 0.0170** |
| TransXNet-GGG-noMCA | 0.9503 ± 0.0019 | 0.8660 ± 0.0069 | 0.8325 ± 0.0013 | 0.8610 ± 0.0016 |
| TransXNet-GGG-MCA | 0.9483 ± 0.0006 | 0.8726 ± 0.0046 | 0.8503 ± 0.0123 | 0.8790 ± 0.0140 |

`TransXNet-GGG-MCA` ablation row, which corresponds to UL-TransXNet, is close but not best on TN5000 alone. This distinction is important because it prevents the paper from over-attributing all gains to the final GGG-MCA setting.

The structure ablation shows that the TransXNet-family modifications are useful but metric-dependent. The G and GG variants improve thresholded operating-point behavior relative to the baseline. The GGG-MCA configuration is not the TN5000 AUC winner, but it remains competitive and

Table 8: Validation-performance audit for the TransXNet-family selection. Values are averages over three TN5000 seeds or five BUSI/AUL training folds. The selected row is the final UL-TransXNet configuration used in the main manuscript tables.

| Variant | TN5000 val AUC | BUSI val AUC | AUL val AUC | Mean val AUC | Selected |
|---|---|---|---|---|---|
| TransXNet | 0.9577 | 0.7796 | 0.7464 | 0.8279 | no |
| TransXNet-GG | 0.9632 | 0.8011 | 0.7390 | 0.8345 | no |
| GGG-noMCA | **0.9700** | 0.8957 | 0.8798 | 0.9152 | no |
| UL-TransXNet | 0.9638 | **0.9199** | **0.8827** | **0.9222** | yes |

corresponds to the best multi-dataset in-domain pattern in this result set. Table 8 audits the model-family choice using validation AUC from the completed runs. The selection rule is to choose the TransXNet-family row with the highest mean validation AUC over TN5000, BUSI, and AUL under the dataset-specific training protocols; the source-controlled model-selection protocol records the contributing CSV files and checksums. UL-TransXNet gives the highest mean validation AUC across TN5000, BUSI, and AUL, while `GGG-noMCA` remains the strongest TN5000 validation-AUC row. Table 9 then shows the corresponding test-set behavior: `TransXNet-GG` is the strongest TN5000-only structure, whereas UL-TransXNet is substantially stronger on BUSI and AUL and therefore gives the better three-dataset average.

### *5.3. TN5000 core-module ablation*

Table 10 isolates the core-module path under the same TN5000 protocol. Differential attention provides the clearest single-module operating-point improvement: `GGG-core+DA` improves balanced accuracy, macro F1, and accuracy over `GGG-core`. MUDD alone gives a smaller gain in thresholded metrics and does not improve AUC in this setting. Adding MCA to the full GGG

Table 9: TN5000-only optimum versus the final multi-dataset configuration. The final reported model is retained for three-dataset behavior, not for the single best TN5000 ablation row.

| Dataset | TransXNet-GG | | UL-TransXNet | |
|---|---|---|---|---|
| | AUC | BalAcc | AUC | BalAcc |
| TN5000 | **0.9517** | **0.8784** | 0.9483 | 0.8726 |
| BUSI | 0.7707 | 0.7000 | **0.8998** | **0.7961** |
| AUL | 0.6972 | 0.6550 | **0.8767** | **0.8200** |
| Mean | 0.8065 | 0.7445 | **0.9083** | **0.8296** |

Table 10: Unified-protocol core-module ablation on TN5000. Mean and standard deviation are computed over three seeds.

| Variant | AUC | BalAcc | F1-macro | Acc |
|---|---|---|---|---|
| GGG-core | 0.9496 ± 0.0003 | 0.8704 ± 0.0083 | 0.8324 ± 0.0136 | 0.8593 ± 0.0133 |
| GGG-core+MUDD | 0.9470 ± 0.0006 | 0.8712 ± 0.0045 | 0.8406 ± 0.0097 | 0.8687 ± 0.0111 |
| GGG-core+DA | 0.9489 ± 0.0016 | **0.8751 ± 0.0044** | 0.8490 ± 0.0100 | 0.8770 ± 0.0099 |
| GGG-core+MUDD+DA-noMCA | **0.9503 ± 0.0019** | 0.8660 ± 0.0069 | 0.8325 ± 0.0013 | 0.8610 ± 0.0016 |
| GGG-core+MUDD+DA+MCA | 0.9483 ± 0.0006 | 0.8726 ± 0.0046 | **0.8503 ± 0.0123** | **0.8790 ± 0.0140** |

configuration lowers TN5000 AUC slightly relative to the no-MCA full model, but improves balanced accuracy, macro F1, and accuracy. This supports a restrained interpretation of MCA: it is useful for decision-level robustness and multi-dataset behavior, not a universal AUC booster.

### *5.4. MCA effect across datasets*

Table 11 compares the final GGG model with and without MCA across the three main datasets. The result explains why the final manuscript should

Table 11: Effect of enabling MCA in the final GGG configuration. Mean and standard deviation are computed over TN5000 seeds or BUSI/AUL folds.

| Dataset | Variant | AUC | BalAcc | F1-macro | Acc |
|---|---|---|---|---|---|
| TN5000 | GGG-noMCA | **0.9503 ± 0.0019** | 0.8660 ± 0.0069 | 0.8325 ± 0.0013 | 0.8610 ± 0.0016 |
| TN5000 | GGG-MCA | 0.9483 ± 0.0006 | **0.8726 ± 0.0046** | **0.8503 ± 0.0123** | **0.8790 ± 0.0140** |
| BUSI | GGG-noMCA | 0.8588 ± 0.0137 | 0.7551 ± 0.0295 | 0.7526 ± 0.0300 | 0.7628 ± 0.0288 |
| BUSI | GGG-MCA | **0.8998 ± 0.0116** | **0.7961 ± 0.0104** | **0.8018 ± 0.0119** | **0.8140 ± 0.0130** |
| AUL | GGG-noMCA | 0.8720 ± 0.0046 | 0.8181 ± 0.0338 | 0.7462 ± 0.0283 | 0.7685 ± 0.0321 |
| AUL | GGG-MCA | **0.8767 ± 0.0009** | **0.8200 ± 0.0487** | 0.7462 ± 0.0405 | 0.7685 ± 0.0418 |

keep MCA in the model family but avoid overclaiming it. On TN5000, MCA improves balanced accuracy, macro F1, and accuracy while slightly reducing AUC. On BUSI, MCA provides a large gain across all reported metrics. On AUL, MCA gives a small AUC and balanced-accuracy gain with nearly unchanged macro F1 and accuracy. Thus, MCA is best described as a module that improves multi-dataset operating behavior, especially on BUSI, rather than as a universally dominant change.

### *5.5. Automatic ROI localization and closed-loop classification*

The main ROI protocol isolates classifier behavior using annotation-derived crops. Figure 4, Table 12, Table 13, and Table 14 evaluate a more complete inference chain: an automatic detector first predicts the lesion box, an expanded ROI is constructed from that prediction, and the same trained UL-TransXNet classifiers are then applied without retraining. The TN5000 automatic-ROI sanity check confirms that the primary benchmark can also be connected to a detector front end. For BUSI and AUL, the detector does not miss any test image in this evaluation, but localization quality differs by

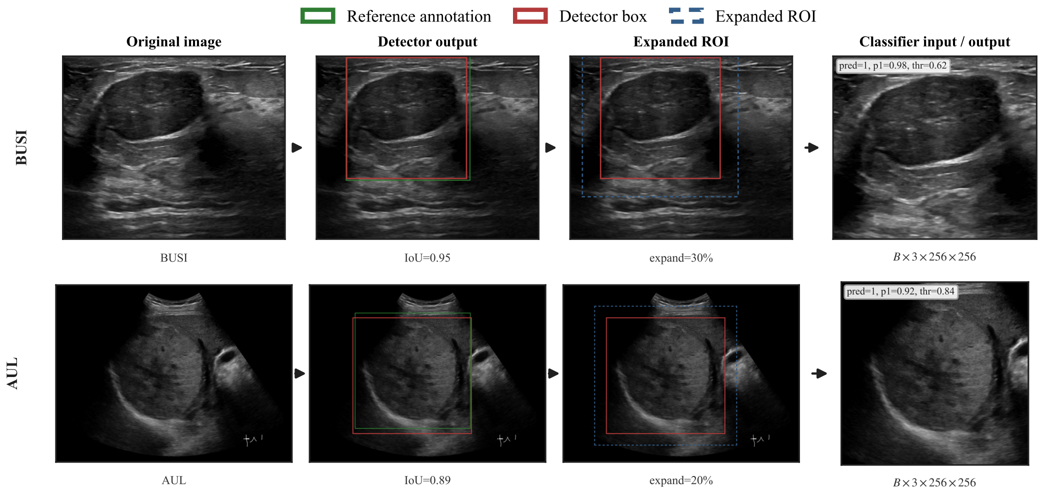


Figure 4: Automatic ROI inference workflow for the BUSI/AUL closed-loop evaluation. Green boxes indicate reference annotations used only for evaluation and visualization; red boxes are detector-predicted lesion boxes; blue dashed boxes are the expanded ROIs used as classifier inputs.

dataset. BUSI reaches a mean IoU of 0.7365, whereas AUL reaches 0.5755 and has a lower high-IoU recall.

Automatic ROI classification is consistently stronger than full-image classification. On BUSI, the automatic ROI improves AUC by 0.0988 and balanced accuracy by 0.0857 over the full-image input. On AUL, the corresponding gains are 0.1066 AUC and 0.0789 balanced accuracy. The automatic ROI remains below oracle ROI, especially on AUL, which is consistent with the lower localization quality in Table 13. These results support the practical use of an automatic ROI front end, while keeping the detector itself outside the main architectural contribution.

Table 12: TN5000 automatic ROI sanity check using existing detector and classifier outputs. Detector values summarize three YOLO11n detector seeds; classification values summarize automatic-ROI closed-loop evaluation across the three detector seeds. No classifier retraining is involved.

| Block | Metric | Value |
|---|---|---|
| Detector | mAP50 | $0.9520 \pm 0.0090$ |
| Detector | mean IoU | $0.7855 \pm 0.0033$ |
| Detector | R@0.75 | $0.7880 \pm 0.0122$ |
| Detector | no-detection rate | $0.0000 \pm 0.0000$ |
| Auto ROI classifier | AUC | $0.9437 \pm 0.0042$ |
| Auto ROI classifier | BalAcc | $0.8686 \pm 0.0056$ |
| Auto ROI classifier | F1-macro | $0.8519 \pm 0.0116$ |
| Auto ROI classifier | Acc | $0.8820 \pm 0.0115$ |

### *5.6. Qualitative Grad-CAM sanity check*

Figure 5 provides a qualitative Grad-CAM sanity check using one trained UL-TransXNet checkpoint per dataset. The maps are generated from late spatial backbone blocks selected for readable ROI-level maps. The activations broadly fall on lesion tissue and adjacent ultrasound context inside the ROI. This visualization is used only as a qualitative sanity check that the classifier is not driven solely by image borders or unrelated background. It is not used as quantitative evidence of causal attention or as a substitute for the ablation and closed-loop ROI experiments above.

Table 13: Detector localization quality used for automatic ROI construction. Values are mean ± standard deviation across five detector folds.

| Dataset | Mean IoU | R@0.50 | R@0.75 | No det. |
|---|---|---|---|---|
| BUSI | 0.7365 ± 0.0241 | 0.8403 ± 0.0141 | 0.6822 ± 0.0548 | 0.0000 |
| AUL | 0.5755 ± 0.0221 | 0.6740 ± 0.0419 | 0.4331 ± 0.0201 | 0.0000 |

Table 14: Closed-loop classification with oracle ROI, automatic detector ROI, and full-image input. Values are mean ± standard deviation across five folds. Bracketed intervals are fold-bootstrap 95% confidence intervals for AUC and balanced accuracy.

| Dataset | Input | AUC | BalAcc | F1-macro | Acc |
|---|---|---|---|---|---|
| BUSI | oracle | 0.8998 ± 0.0116 [0.8880, 0.9067] | 0.7961 ± 0.0104 [0.7865, 0.8045] | 0.8018 ± 0.0119 | 0.8140 ± 0.0130 |
| BUSI | auto | 0.8725 ± 0.0158 [0.8564, 0.8845] | 0.7839 ± 0.0179 [0.7704, 0.8010] | 0.7893 ± 0.0176 | 0.8031 ± 0.0167 |
| BUSI | full | 0.7737 ± 0.0044 [0.7701, 0.7777] | 0.6982 ± 0.0237 [0.6762, 0.7178] | 0.6915 ± 0.0378 | 0.7023 ± 0.0418 |
| AUL | oracle | 0.8767 ± 0.0009 [0.8759, 0.8776] | 0.8200 ± 0.0487 [0.7755, 0.8599] | 0.7462 ± 0.0405 | 0.7685 ± 0.0418 |
| AUL | auto | 0.8507 ± 0.0222 [0.8320, 0.8708] | 0.7593 ± 0.0288 [0.7340, 0.7847] | 0.6908 ± 0.0561 | 0.7165 ± 0.0713 |
| AUL | full | 0.7441 ± 0.0111 [0.7344, 0.7538] | 0.6805 ± 0.0214 [0.6623, 0.6999] | 0.6787 ± 0.0203 | 0.7512 ± 0.0190 |

### *5.7. Uncertainty and calibration analysis*

Table 15 reports bootstrap confidence intervals and calibration metrics for UL-TransXNet using the same seed/fold-level reporting protocol as the main tables. The confidence intervals provide a compact statistical view of the repeated-run stability, while ECE and Brier score quantify probability calibration after the validation-derived temperature-scaling step.

### *5.8. Efficiency and mobile prototype*

The Android experiment evaluates a TN5000-calibrated prototype of UL-TransXNet. Table 16 and Figure 6 show a trade-off among three inference variants on one device. A single checkpoint provides the lowest end-to-end

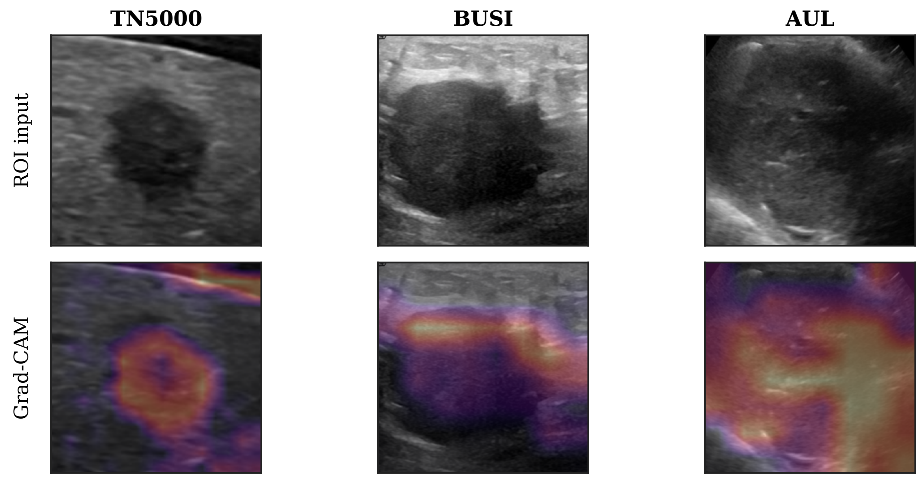


Figure 5: Grad-CAM sanity-check visualization for representative correctly classified ROI inputs. Columns correspond to datasets; the upper row shows the ROI input and the lower row shows the Grad-CAM overlay. The maps are generated from a single UL-TransXNet checkpoint for each dataset and are intended as qualitative localization checks only.

latency at 116.5 ms per image. Adding horizontal-flip TTA improves both AUC and balanced accuracy, reaching 0.9466 AUC and 0.8762 balanced accuracy at 226.2 ms per image. The three-checkpoint ensemble further increases AUC to 0.9485, but it reduces balanced accuracy at the selected operating threshold and raises latency to 738.6 ms per image. Thus, the single and TTA variants are the practical interactive settings for this prototype, whereas the ensemble is better interpreted as an offline ranking or review option. These measurements are not a full mobile-baseline benchmark.

## 6. Discussion

The results support a restrained conclusion. UL-TransXNet is a competitive compact ultrasound classifier with favorable multi-dataset in-domain be-

Table 15: Uncertainty and calibration summary for UL-TransXNet on the main test sets. Values follow the manuscript seed/fold-level reporting protocol; intervals are bootstrap 95% confidence intervals over seeds or folds. ECE uses 10 confidence bins.

| Dataset | AUC 95% CI | BalAcc 95% CI | ECE | Brier |
|---|---|---|---|---|
| TN5000 | 0.9483 [0.9475, 0.9488] | 0.8726 [0.8662, 0.8771] | 0.0220 | 0.0741 |
| BUSI | 0.8998 [0.8880, 0.9067] | 0.7961 [0.7865, 0.8045] | 0.0696 | 0.1358 |
| AUL | 0.8767 [0.8759, 0.8776] | 0.8200 [0.7755, 0.8599] | 0.1283 | 0.1597 |

Table 16: Android prototype results on the TN5000 test set. Latency includes ROI preprocessing and ONNX Runtime inference.

| Mode | Runs | AUC | BalAcc | Avg ms | P95 ms |
|---|---|---|---|---|---|
| Single | 4 | 0.9402 | 0.8688 | 116.5 | 129.0 |
| + HFlip TTA | 2 | 0.9466 | **0.8762** | 226.2 | 246.6 |
| 3-ckpt ensemble + TTA | 2 | **0.9485** | 0.8723 | 738.6 | 763.5 |

havior, but it should not be described as uniformly superior on every dataset and every metric. On TN5000, ConvNeXt-Tiny remains slightly stronger in the repeated-seed table; on AUL, Swin-T slightly exceeds UL-TransXNet in macro F1 and raw accuracy despite lower AUC and balanced accuracy. The proposed model becomes clearly advantageous in the multi-dataset view: it is substantially stronger on BUSI, has higher AUC and balanced accuracy on AUL, and has the best average performance across the three public benchmarks under dataset-specific training.

The TN5000 ablations clarify the role of the architecture components. The structure-level progression shows that `TransXNet-GG` is the strongest

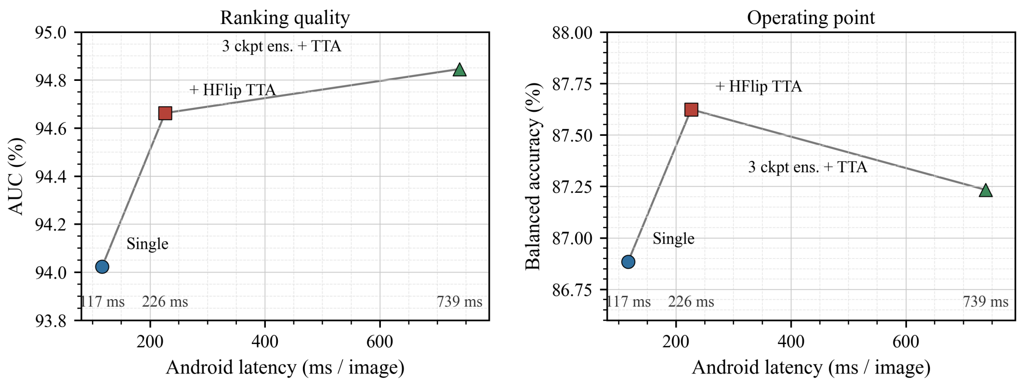


Figure 6: Accuracy–latency trade-off of the Android prototype variants on the TN5000 test set.

TN5000 variant, while the `TransXNet-GGG-MCA` configuration used by UL-TransXNet is the best three-dataset choice in the validation audit and final result tables. The module-level ablation shows that differential attention is the clearest contributor to TN5000 thresholded metrics. MUDD has a smaller and less uniform effect. MCA does not uniformly improve TN5000 AUC, but it improves TN5000 F1 and accuracy and produces large BUSI gains. This means that MCA should be framed as an operating-point and multi-dataset behavior module rather than as the sole reason for the model family performance.

This interpretation is important in medical image analysis. Ultrasound datasets differ substantially in organ anatomy, acquisition quality, lesion size, and class prevalence. A small AUC difference on one thyroid benchmark is less informative than the pattern across several public benchmarks and operating points. The evidence suggests that the proposed architecture is useful because it gives a favorable balance across dataset-specific evaluations, not because every module monotonically improves every metric.

The automatic ROI experiments address a separate practical concern. The main classifier is trained under an annotation-derived ROI protocol, but the TN5000, BUSI, and AUL closed-loop evaluations show that detector-predicted crops can replace oracle crops with a moderate performance loss and remain clearly stronger than full-image classification where that comparison is evaluated. This result should be interpreted as workflow evidence rather than detector novelty. In particular, AUL localization is weaker than BUSI localization, which explains part of the remaining gap between automatic and oracle ROI classification.

The Android results remain a practical feasibility check. Many ultrasound classification papers stop at workstation-side accuracy. The mobile experiment shows that a TN5000-calibrated prototype can run in a low-latency mode while preserving strong classification quality, and that a more expensive ensemble mode is available when latency is less critical. These measurements do not constitute clinical deployment validation or prove superiority over all mobile baselines, but they show that UL-TransXNet is compatible with edge-side inference constraints.

*Limitations*

This study has several limitations. First, the evaluation is still based on public retrospective datasets, and label definitions, class proportions, and preprocessing conditions are not fully uniform across datasets. BUSI and AUL are dataset-specific in-domain evaluations rather than thyroid-to-breast or thyroid-to-liver external generalization tests. Second, the BUSI/AUL protocol uses five training folds evaluated on a fixed held-out test set, so the reported standard deviations reflect training-fold variation rather than an outer cross-validation estimate. Third, the work focuses on binary benign–malignant classification rather than detection, segmentation, or multi-class clinical diagnosis. Fourth, the automatic ROI experiment uses a practical one-class detector only to form the classifier input; it does not establish a new lesion-detection method, and the AUL results show that localization quality can still limit closed-loop performance. Fifth, the mobile experiment demonstrates prototype feasibility but not a full clinical workflow under realistic operator interaction and acquisition variability. Finally, external robustness remains an open problem because the evaluation is still limited to public retrospective datasets rather than prospective multi-center cohorts. Future studies should therefore focus on broader external validation, stronger detector-classifier integration, calibration under distribution shift, and clinically grounded prospective evaluation.

## 7. Conclusion

This paper presents UL-TransXNet, a compact ROI-based TransXNet-family architecture for benign–malignant ultrasound lesion classification. The evidence supports a restrained but coherent conclusion. UL-TransXNet is competitive on TN5000, clearly stronger on BUSI and AUL, and strongest on average across the three public datasets under dataset-specific training and the unified reporting protocol. TN5000 ablations show that the architecture gains are not explained by a single module: the GG structure is the strongest thyroid-only variant, differential attention improves thresholded behavior, and MCA mainly improves multi-dataset operating behavior.

Automatic ROI experiments further show that detector-predicted crops can connect the classifier to a practical localization front end and remain stronger than full-image classification, including a TN5000 automatic-ROI sanity check for the primary benchmark. Android experiments show that the model can support prototype speed–accuracy trade-offs on mobile hardware. At the same time, broad external generalization and clinical deployment remain unresolved because all reported datasets are public retrospective benchmarks. Future studies should therefore focus on larger external validation, stronger automatic ROI generation, more rigorous calibration under distribution shift, and deployment evaluation under realistic clinical workflows.

## Appendix A. Supplementary Benchmark Tables

Table A.17 reports the full benchmark table used to support the main comparison. Values are mean ± standard deviation across TN5000 seeds or BUSI/AUL training folds evaluated on fixed held-out test sets. The proposed model is UL-TransXNet, which corresponds to the TransXNet-GGG-MCA configuration in the ablation notation; baseline rows are taken from the same comparison protocol.

## Appendix B. Reproducibility Note

All main benchmark tables were generated from the seed/fold summary CSV files used by the experiment pipeline. The project archive records the dataset split manifests, ROI construction scripts, training entry points, aggregation scripts, model checkpoints, Android artifacts, validation thresh-

Table A.17: Full benchmark across TN5000, BUSI, and AUL using the unified reporting convention. Best per column is shown in bold.

| Method | TN5000 AUC | TN5000 BalAcc | BUSI AUC | BUSI BalAcc | AUL AUC | AUL BalAcc |
|---|---|---|---|---|---|---|
| UL-TransXNet | 0.9483 ± 0.0006 | 0.8726 ± 0.0046 | **0.8998 ± 0.0116** | **0.7961 ± 0.0104** | **0.8767 ± 0.0009** | **0.8200 ± 0.0487** |
| ResNet50 | 0.6531 ± 0.0081 | 0.6319 ± 0.0067 | 0.7519 ± 0.0332 | 0.6826 ± 0.0205 | 0.5481 ± 0.0771 | 0.5426 ± 0.0343 |
| EfficientNet-B0 | 0.7856 ± 0.0242 | 0.7026 ± 0.0165 | 0.7659 ± 0.0368 | 0.7061 ± 0.0317 | 0.6099 ± 0.1220 | 0.5958 ± 0.0912 |
| MobileNetV3-Large | 0.8282 ± 0.0083 | 0.7626 ± 0.0082 | 0.7947 ± 0.0547 | 0.7311 ± 0.0462 | 0.7357 ± 0.0062 | 0.6611 ± 0.0161 |
| Swin-T | 0.9497 ± 0.0025 | 0.8782 ± 0.0051 | 0.8190 ± 0.0091 | 0.7137 ± 0.0095 | 0.8319 ± 0.0117 | 0.7783 ± 0.0325 |
| DenseNet121 | 0.9360 ± 0.0062 | 0.8609 ± 0.0074 | 0.8154 ± 0.0278 | 0.7291 ± 0.0471 | 0.6729 ± 0.0358 | 0.6304 ± 0.0706 |
| ConvNeXt-Tiny | **0.9508 ± 0.0020** | **0.8794 ± 0.0023** | 0.8050 ± 0.0237 | 0.7261 ± 0.0420 | 0.8274 ± 0.0284 | 0.6898 ± 0.0248 |
| RepViT-M1.1 | 0.8403 ± 0.0510 | 0.7603 ± 0.0427 | 0.8028 ± 0.0134 | 0.7137 ± 0.0442 | 0.6797 ± 0.0029 | 0.6320 ± 0.0225 |
| EfficientFormer-L1 | 0.9249 ± 0.0021 | 0.8484 ± 0.0088 | 0.7917 ± 0.0342 | 0.7213 ± 0.0226 | 0.7196 ± 0.0136 | 0.6293 ± 0.0544 |

olds, fitted temperatures, prediction CSV files, and a table-level source index required to reproduce the reported TN5000, BUSI, AUL, automatic-ROI, and Android prototype results. The prepared source-controlled release is intended to include the scripts, compact configurations, split metadata, table summaries, prediction-level CSV files, and the model-selection protocol with source-file checksums that can be shared without redistributing dataset images or large trained weights.

## CRediT authorship contribution statement

Zihan Li: Conceptualization, methodology, software, experiments, and writing—original draft.
Cong Yang: Supervision, conceptualization, writing—review and editing, resources, and project administration.

## Ethics statement

This study is a secondary analysis of publicly available, de-identified ultrasound datasets. No new patient data were collected by the authors. Ethical approval and consent procedures are governed by the original dataset publications or data providers; therefore, no additional institutional review board approval was required for this retrospective computational study.

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### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

The datasets analyzed in this study are publicly available from their original sources, which are cited in the manuscript. A reproducibility package is prepared for release at https://github.com/Afr1ste/ul-transxnet and includes source code, split/protocol configuration files, environment specifications, paper-generation scripts, Android demonstration source, artifact-validation utilities, compact result summaries, table provenance, validation thresholds, temperature-scaling values, and prediction CSV files required for table regeneration. Dataset image files, generated ROI image folders, trained checkpoints, detector weights, ONNX binaries, Android application packages, and full raw run logs are not redistributed in the repository. These restricted or large artifacts are retained in the project archive and may be made available from the corresponding author upon reasonable request where permitted by the applicable terms of the original datasets and software providers.

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